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SPPU M.Sc. Data Science — Free Study Material

Python Programming for Data Science

DS-553-MJ

Sample / Practice Question Paper

Semester II · Core · Practice Paper

& This is a SAMPLE / PRACTICE question paper prepared for revision from the SPPU M.Sc. Data Science syllabus. It is NOT an official or past university examination paper. Use it only to test your preparation.

Time: 3 Hours

Maximum Marks: 35

Instructions: (1) All questions carry equal weightage where indicated. (2) Figures to the right indicate full marks. (3) Draw neat diagrams / write code wherever necessary.

Q1. Answer any FIVE of the following. [5 × 2 = 10]

- a) Differentiate list and tuple.
- b) What is a lambda function?
- c) What is a dictionary?
- d) What is exception handling?
- e) What is a NumPy array?
- f) What is a Pandas DataFrame?
- g) What does groupby do?

Q2. Answer the following. [10]

A) Explain Python data types, conditional statements and loops with examples.
OR

B) Explain lists, tuples, dictionaries and sets with operations.

Q3. Answer the following. [10]

A) Explain functions, lambda, map, filter and reduce with examples.
OR

B) Explain modules, packages, file handling and exception handling.

Q4. Answer the following. [5]

A) Explain NumPy arrays and their operations.
OR

B) Explain Pandas DataFrame, handling missing data and groupby.

